

ANTHONY ZHAI

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EDUCATION

Princeton University

B.A. in Mathematics + Computer Science

Princeton, NJ

September 2023 - May 2027

Relevant Coursework: Data Structures and Algorithms, Intro to Programming Systems, Intro to Theoretical Machine Learning, Reinforcement Learning, Probability and Stochastic Systems

GPA: 3.8/4.0

EXPERIENCE

Databricks

Software Engineer Intern

Mountain View, CA

Jun - Sep 2026

- Designed, implemented, and rolled out a Kubernetes CronJob to replace a legacy workspace garbage collection service, reducing p95 workspace deletion latency by $65\times$, accelerating handling of already-deleted workspace data by $1000\times$, and clearing 20M+ rows of stale + backlogged records across 100k+ cancelled production workspaces per month.
- Improved reliability and operational control through workload isolation, on-demand purging, granular releases and rollback, and a full observability suite with metrics, dashboards, logs, and on-call alerting.

Predigle

Software Engineer Intern

May - Aug 2023, 2024, 2025

- Architected and developed a full-stack geospatial web application (Angular, Leaflet, Django) for visualizing multi-layer data with dynamic toggling, shape-drawing, and real-time layer generation — initially applied to optimize coconut supply chain logistics, and now being productized for broader use cases such as visualizing localized market performance, competitor locations, and customer proximity.
- Co-developed a LangChain-based Retrieval-Augmented Generation (RAG) chatbot using open-source LLMs to help automate prior authorization assessment, enabling healthcare teams to query and summarize complex medical PDFs with higher accuracy and efficiency for prescription fulfillment.
- Built ML and LLM/OCR pipelines for insurance claim overpayment classification, achieving 97% accuracy on 400k+ claim records, identifying \$250k+ in new claims, and unlocking 150k+ previously unlabeled scanned-document data points.

Princeton University Power Electronics Research Lab

Machine Learning Research Intern

Jun - Aug 2022

- Developed a novel neural network to model magnetic core loss that uses the Fast Fourier Transform algorithm to more efficiently encode time-series data, using $<20\%$ the parameters of LSTM networks with $<15\%$ training time and equal performance.
- Analyzed downsampling methods for frequency and peak flux density to assess the relative significance of variables in predicting core loss and to reduce the parameter search space by 20%.

PROJECTS

CGBNet: A Deep Learning Framework for Compost Classification *Python, Tensorflow, Keras, Computer Vision, Deep Learning, Transfer Learning*

Co-first authored research article for CGBNet—a deep learning framework using computer vision to classify nitrogen-rich and carbon-rich compost for automating composting—published in IEEE Access, an engineering application journal (3.9 impact factor and 30% acceptance rate).

Visionary *Python, Django, Django-Rest-Framework, Flutter, Dart, Tensorflow, Keras, NLP*

Developed the Visionary mobile app to help the visually impaired interact with the world—generates narrated descriptions of the user's surroundings using text OCR, object detection, and NLP algorithms.

SKILLS

Programming Languages: Python, Java, C++, Javascript, Dart, HTML / CSS, SQL

Libraries: Tensorflow, PyTorch, Numpy, Pandas, Flutter, React, Node, Vue, Flask, Django, OpenCV

Technologies: Git, Bash, AWS, GCP, Docker, PostgreSQL, Firebase, Figma, NPM, UNIX

AWARDS

Most Innovative Startup Idea

HackPrinceton

Created EduLecture.ai, a platform for generating personalized lecture videos using only Google Slide presentations. Built with a React.js frontend and a Rest API backend powered by voice-generative AI tools, EduLecture.ai bridges the accessibility gap for remote education.